

Nutrition retake examination for medical students

1. What is the energy density?
 - a) The number of calories present in each gram of food
 - b) The number of vitamins and minerals in the food
 - c) The ratio of protein to carbohydrate in the food
 - d) The total volume of food relative to its weight**Answer: a)**
2. Which of the following is NOT a basic principle of diet planning?
 - a) Adequacy
 - b) Balance
 - c) Moderation
 - d) Elimination of food groups**Answer: d)**
3. A 40-year-old man develops fat malabsorption after gastric bypass surgery. Which clinical sign is more likely?
 - a) Weight gain
 - b) Steatorrhea and deficiency of fat-soluble vitamins
 - c) Hypercalcemia
 - d) Peripheral edema**Answer: b)**
4. A teenager with a diet containing soda and fast food has decreased bone density. What is the main cause?
 - a) High protein intake
 - b) Deficiency of calcium and vitamin D
 - c) Excessive fiber intake
 - d) Iron deficiency**Answer: b)**
5. Which of the following options is most likely to cause a slower glycemic response?
 - a) White wheat bread
 - b) White sugar
 - c) Whole oats
 - d) Glucose solution**Answer: c)**
6. Which of the following dietary changes is more effective in reducing the risk of cardiovascular diseases related to carbohydrate consumption?
 - a) Reducing all carbohydrates
 - b) Replacing refined grains with whole grains
 - c) Limiting added sugars and increasing fiber intake
 - d) Eliminating fruit from the diet**Answer: c)**

7. During prolonged fasting, which sequence of metabolic changes occurs?

- a) Glycogen breakdown → Gluconeogenesis → Ketone body formation
- b) Ketone body formation → Glycogen breakdown → Glycolysis
- c) Glucose oxidation → Insulin secretion → Lipolysis
- d) Lipogenesis → Protein synthesis → Glycolysis

Answer: a)

8. What is the biochemical basis of symptoms in lactose intolerance?

- a) Overproduction of the enzyme lactase
- b) Absorption of lactose into the bloodstream without digestion
- c) Deficiency of lactase enzyme leading to fermentation of undigested lactose in the large intestine
- d) Increased insulin production

Answer: c)

9. Which change in dietary fats has the greatest effect on lowering LDL cholesterol levels?

- a) Replacing monounsaturated fats (MUFA) with saturated fats (SFA)
- b) Replacing trans fats with polyunsaturated fats (PUFA)
- c) Increasing dietary cholesterol intake
- d) Eliminating omega-3 fatty acids

Answer: b)

10. A food product that remains semi-solid at room temperature and contains trans fats indicates what about its production process?

- a) It contains essential fatty acids
- b) It is fully hydrogenated
- c) It is partially hydrogenated to increase shelf life
- d) It is enriched with omega-3 fats

Answer: c)

11. How does excessive intake of saturated fats affect the lipid profile and the risk of cardiovascular diseases?

- a) Increases HDL and decreases LDL
- b) Decreases triglycerides and increases HDL
- c) Increases LDL and contributes to atherosclerosis
- d) Has no effect on lipid profile

Answer: c)

12. During fasting, fat becomes the main energy source for the body. Which characteristic of fats explains their efficiency as long-term energy storage?

- a) High energy content (9 kcal per gram)
- b) Rapid digestibility
- c) High water content
- d) Immediate conversion to glucose

Answer: a)

- 13.** A patient with a normal BMI but decreased muscle mass visits. How do you expect their resting energy expenditure (REE) to compare to a healthy individual?
- a) Higher
 - b) Lower
 - c) The same
 - d) Unpredictable

Answer: b)

- 14.** A patient with a history of a strict diet and rapid weight loss has experienced a decrease in basal metabolic rate. What is the main cause of this phenomenon?
- a) Increased fat mass
 - b) Decreased fat-free mass
 - c) Increased physical activity
 - d) High protein intake

Answer: b)

- 15.** A patient with a history of high carbohydrate intake and obesity visits. If their respiratory quotient (RQ) is greater than 1, which metabolic state is more likely?
- a) Ketosis
 - b) Lipogenesis (conversion of carbohydrate to fat)
 - c) Gluconeogenesis
 - d) Increased protein consumption

Answer: b)

- 16.** A 45-year-old man with a history of hyperthyroidism has lost weight. Which mechanism plays the biggest role in this condition?
- a) Increased insulin secretion
 - b) Increased basal energy expenditure
 - c) Decreased physical activity
 - d) Increased fat storage

Answer: b)

- 17.** PKU is characterized by absence or deficiency of:
- a) serotonin
 - b) tyrosine aminotransferase
 - c) tryptophan
 - d) phenylalanine hydroxylase

answer: d)

18. What percentage of protein synthesis is derived from dietary amino acid intake?

- a) Approximately 10%
- b) Approximately 20%
- c) Approximately 50%
- d) Approximately 75%

answer: b

19. What are the most negative effects of long-term consumption of high protein diets?

- a) Effects on digestion and gut health
- b) decreased muscle growth
- c) increased risk of kidney damage
- d) Improved metabolic rate

answer c

20. Which amino acid is particularly low in cereal proteins?

- a) Lysine
- b) Methionine
- c) Leucine
- d) Tryptophan

Answer a

21. Which of the following is considered a core anthropometric measurement?

- a) Assessment of dental health
- b) Evaluation of vision acuity
- c) Measurement of body length or height
- d) Quantification of hearing capacity

Answer c

22. According to established WHO standards, a Z-score for head circumference falling within the range of -2 to +2 is considered:

- a) Abnormally high
- b) Pathologically low
- c) Within normal limits
- d) Requiring immediate medical intervention

Answer c

23. Kwashiorkor is a form of severe deficiency in what nutrient?

- a) Sugar
- b) Protein
- c) Vitamins
- d) Water

Answer b

24. A key sign of Kwashiorkor is:

- a) Severe wasting
- b) Monkey face
- c) Edema (swelling)
- d) High energy levels

Answer c

25. What does the term "Marasmus" relate to a lack of?

- a) Energy malnutrition
- b) Protein malnutrition
- c) Vitamin deficiency
- d) Mineral deficiency

Answer a

26. What is the common appearance of a person with Marasmus?

- a) Swollen body
- b) Severely wasted
- c) Overweight
- d) Normal weight

Answer b

27. What is another name for Vitamin A?

- a) Ascorbic acid
- b) Thiamin
- c) Retinol
- d) Biotin

Answer c

28. A deficiency of Vitamin A can lead to which condition?

- a) Rickets
- b) Scurvy
- c) Pellagra
- d) Night blindness

Answer d

29. A daily exposure of 6000 RAEs of Vitamin A from supplements can be linked to:

- a) A stronger immune system
- b) Fetal malformations
- c) Improved eyesight
- d) Higher metabolism

Answer b

30. What is another name for Vitamin D?

- a) Retinol
- b) Cholecalciferol
- c) Niacin
- d) Pyridoxine

Answer b

31. A deficiency of Vitamin D can lead to which bone disease in children?

- a) Osteoporosis
- b) Pellagra
- c) Rickets
- d) Scurvy

Answer c

32. What is a sign of Vitamin E deficiency?

- a) Night blindness
- b) Scurvy
- c) Peripheral neuropathy
- d) Rickets

Answer c

33. What is the main function of Vitamin K?

- a) Bone growth
- b) Muscle building
- c) Blood clotting
- d) Eye health

Answer c

34. A deficiency of Vitamin C can lead to what disease?

- a) Rickets
- b) Night blindness
- c) Scurvy
- d) Beriberi

Answer c

35. What are the two types of dietary iron?

- a) Soluble and insoluble
- b) Organic and inorganic
- c) Heme and nonheme
- d) Fast and slow

Answer c

36. A lack of iron can cause what condition?

- a) Anemia
- b) Pellagra
- c) Rickets
- d) Beriberi

Answer a

37. Zinc is needed for the activity of hundreds of:

- a) Hormones
- b) Enzymes
- c) Red blood cells
- d) Vitamins

Answer b

38. Iodine is a key part of which hormones?

- a) Thyroid hormones
- b) Insulin
- c) Growth hormones
- d) Adrenaline

Answer a

39. The earliest clinical sign of iodine deficiency is usually:

- a) Anemia
- b) Goiter
- c) Night blindness
- d) Numbness

Answer b

40. Feeding through GI has been shown to:

- a) Attenuate the catabolic response
- b) Lose immunologic function
- c) More infectious complications
- d) More septic morbidity

Answer a